**STUDYNEST | CLASS 11 PHYSICS**

# MARKING SCHEME AND MODEL ANSWERS

## Chapter 7: Gravitation

**Maximum Marks: 40 | Suggested time: 60 minutes**

Use after attempting the paper. Accept equivalent correct methods.

### Suggested mark allocation:

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

**Q1 - 1 mark**

The radius vector sweeps equal areas in equal times.

**Q2 - 1 mark**

F=Gm_1m_2/r^2.

**Q3 - 1 mark**

N m^2 kg^-2.

**Q4 - 1 mark**

g=GM/R^2.

**Q5 - 1 mark**

U=-GMm/r.

**Q6 - 1 mark**

No.

**Q7 - 1 mark**

Gravity.

**Q8 - 1 mark**

T^2 proportional to r^3.

**Q9 - 2 marks**

4.00x10^-10 N.

**Q10 - 2 marks**

490 N.

**Q11 - 2 marks**

1/sqrt2.

**Q12 - 2 marks**

1/4.

**Q13 - 2 marks**

DeltaU=mgh.

**Q14 - 2 marks**

8.

**Q15 - 3 marks**

g_h/g=(2/3)^2=4/9.

**Q16 - 3 marks**

sqrt(2gR)=4.47x10^3 m/s.

**Q17 - 3 marks**

sqrt(GM/r)approx 7.56x10^3 m/s.

**Q18 - 3 marks**

150 N.

**Q19 - 4 marks**

GMm/r^2=mv^2/r v=sqrt(GM/r); T=2pir/v=2pisqrt(r^3/GM).

**Q20 - 4 marks**

vapprox 7.56x10^3 m/s; E=-GMm/(2r)approx -2.86x10^10 J.